

Multiplexed microsatellite markers for genetic studies of beech

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Abstract

European beech (*Fagus sylvatica* L.) is one of the economically most important broadleaved tree species in Europe and has become a model for studying climate change effects on forests. Multiplex PCR of microsatellites is a fast and cost-effective technique allowing high-throughput genotyping. Here we present the procedure used to develop two multiplex kits (8-plexes) for European beech. We paid particular attention to quality control throughout all steps of the multiplex kits development (null allele detection, error rate measurements, linkage disequilibrium). Preliminary assays suggest that the 16 amplified loci are largely devoid of null alleles and allow rapid and cost-effective genotyping of beech with low error rates. The two kits, which differ in their levels of polymorphism, most likely due to marker origin, were also informative in seven other beech species tested.

Keywords: *Fagus sylvatica*, microsatellites, multiplex PCR, polymorphism, repeat number

Received 17 June 2011; revision received 18 October 2011; accepted 25 October 2011

Introduction

European beech (*Fagus sylvatica* L.) is one of the most widespread broadleaved tree species in Europe, covering about 12 million ha (Muhs & von Wuehlisch 1993) extending from Scandinavia to Sicily and from Spain to Ukraine (von Wuehlisch 2008). Its wide distribution and great economic value have made *F. sylvatica* one of the most thoroughly investigated European tree species, having been the subject of studies in various fields such as ecology, palaeoecology and genetics (Magri *et al.* 2006 and references therein) and an important model species to assess forest responses to climate change (Jump *et al.* 2006a,b, 2008; Kramer *et al.* 2010). It has been the object of numerous genetic surveys using chloroplast DNA markers (Demesure *et al.* 1996; Vettori *et al.* 2004; Magri *et al.* 2006), isozyme markers (Comps *et al.* 2001), AFLP (Jump & Peñuelas 2007) and microsatellite markers (SSRs) (Jump & Peñuelas 2006), providing important information about population dynamics, spatial or temporal genetic structure, migration history and distribution of genetic diversity. SSRs are particularly well suited for such ecological genetics research thanks to their high degree of polymorphism (Streiff *et al.* 1999). However, by surveying recently published SSR studies of beech ($n = 40$), we found that more than half of the studies lack

appropriate quality controls (e.g. null allele detection, error rate measurements, linkage disequilibrium) and none of them relies on multiplex PCR. Therefore, analysing large beech populations at multiple SSR markers remains error-prone, expensive and time-consuming. Multiplexing consists in the amplification of several markers in a single PCR. The technique can significantly improve genotyping throughput as well as cost-effectiveness. Yet, large multiplexing (eight or more markers) is still uncommon because of concerns of long development procedures and complex reaction interactions (Guichoux *et al.* 2011a). In this study, we developed two multiplex kits for population genetics analysis in *F. sylvatica*, each made of eight SSR markers. We describe the complete procedure used to design the two multiplex kits, paying attention to quality controls and potential transferability to other beech species.

Material and methods

Biological material

The biological material originates from four beech stands located in southern France (Table 1). In 2009, 40 or 45 adult trees (pairwise distance between trees <30 m) were sampled in each of the four stands (total $n = 165$ individuals). For each individual, fresh leaves or winter buds were collected and stored in plastic bags with silica gel.

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Table 1 Geographic positions of the four beech (*Fagus sylvatica* L.) stands

Site code	Site name	Number of samples	Latitude	Longitude	Altitude (m asl)
HAY	Hayra	40	43°04'8.16" N	01°21'46.26" W	566
JAU	Jaura	40	43°01'44" N	00°53'43" W	600
JOY	Joyeuse	40	44°45'52" N	00°22'43" W	90
NOH	Nohèdes	45	42°37'8.69" N	02°15'48.8" E	1200

The two multiplex kits were also tested on eight individuals of *Fagus crenata* (Japan), four of *F. grandifolia* (Canada) and 15 individuals coming from the arboretum of Les Barres (France) including six species: *F. orientalis* ($n = 5$), *F. taurica* ($n = 4$), *F. japonica* ($n = 2$), *F. engleriana* ($n = 2$), *F. grandifolia* ($n = 1$) and *F. lucida* ($n = 1$) to test their portability.

DNA isolation

Total nuclear DNA was extracted from a standardized quantity of plant tissue (five leaf discs, $\varnothing = 5$ mm or two leaf buds). Plant material was disrupted using a Mixer Mill MM300 (Retsch, Germany). A 3 mm tungsten bead was added in each well of 96-well plates which were then frozen in liquid nitrogen for 2 min prior to two cycles of 1-min disruption at 25 Hz. For DNA isolation, we used Invisorb DNA plant HTS 96 kit (Invitex, Germany) and followed manufacturer's instructions. DNA concentration was measured on an eight channel Nanodrop spectrophotometer. Each sample was adjusted to a concentration of 10 ng/ μ L using a STARlet 8-channel robot (Hamilton, USA).

Multiplex PCR optimization

Kit-1. We assessed 14 SSR loci originating from microsatellite-enriched cloned banks to develop a first multiplex (8-plex). Five highly validated genomic SSRs had been developed by Pastorelli *et al.* (2003), Vornam *et al.* (2004) and Asuka *et al.* (2004) for *F. sylvatica* or *F. crenata* and nine markers had been developed by G. G. Vendramin (personal communication) for *F. sylvatica*. None of these markers had been multiplexed before. Criteria for SSR inclusion in the first 8-plex were the number of alleles (>3), lack of evidence for null alleles (tested using half-sib families) and good amplification quality.

Kit-2. For the development of a second kit, we used an equimolar mixture of DNA isolated from six individuals. This beech DNA was screened, and microsatellite loci were identified and isolated using 454 GS-FLX Titanium (Roche Applied Science) pyrosequencing of enriched library technology (procedure carried out by Geno-

screen, Lille, France; Malausa *et al.* 2011). DNA was enriched for the following motifs (AG)₁₀, (AC)₁₀, (AAC)₈, (AGG)₈, (ACG)₈, (AAG)₈, (ACAT)₆ and (ATC-T)₆, and the resulting library was sequenced using 1/32th of a 454 run. Among the 227 loci found (QDD 1.3; Megléc *et al.* 2010), putative markers to be included in the 8-plex were retained on the basis of microsatellite motif repeat number (>9). Then, 48 SSRs were assessed and sorted using the same criteria as *kit-1*. Among these, a subset of 17 loci was selected for the next steps towards completion of *kit-2*.

In a first step, all SSRs loci were validated in simplex reactions using the M13-tail technique (Schuelke 2000). Subsequently, PCR products were visualized on a capillary sequencer. Low quality profiles, *that is* markers showing poor amplification, excessive stuttering or off-ladder microvariants (variants differing from the expected periodicity of 2 bp), were excluded from further analyses. To assess the presence of null alleles, 10 families (composed of the female parent and seven offspring) were genotyped at all loci (De Sousa *et al.* 2005). SSRs were also tested with MICRO-CHECKER 2.2.0.3 (Van Oosterhout *et al.* 2004). Once validated in simplex and prior to multiplexing, primers were tested for possible interactions using a local BLAST. We used MULTIPLEX MANAGER 1.2 (Holleley & Geerts 2009) to compute the complementary threshold (the maximum number of AT or CG matches for any two primers within a multiplex reaction) which was set to seven. Further validations, microsatellites scoring and error rate measurements were performed on both multiplex kits.

Polymerase chain reactions were carried out using the Qiagen Multiplex PCR kit (Qiagen, Germany) following the manufacturer instructions. PCRs were performed in a total volume of 13 μ L, consisting of 4 μ L of Qiagen Multiplex Buffer (2 $\times$), 4.75 μ L of sterile water, 1.25 μ L primer premix and 3 μ L template DNA. Concentrations and label dyes for each primer pair in the primer premix are shown in Table 2. Both multiplex kits had the same PCR conditions: samples were denatured at 95 °C for 15 min, and this was followed by 30 cycles consisting of a denaturing step at 94 °C for 30 s, an annealing step at 55 °C for 1 min and an extension step at 72 °C for 1 min. A final extension step at 60 °C for 30 min was added after 30

Table 2 Characteristics of *kit-1* (shaded area) and *kit-2* (blank area), based on 165 samples of *Fagus sylvatica* L. from southern France

Locus	Primer sequences (5'–3')	Reference	Dye	[C]	Motif	Size (bp)	nA	A	H _o	H _e	F _{IS}
csolfagus_31	TCTATTGACACAAAGAATAAGAACACC CTTGGCAAGAAAAGGGGATT	G. G. Vendramin, personal communication	VIC	1.50	(AG) ₁₂	104–126	9	6.25	0.756	0.713	–0.061
sfc_1143	TGGCATCTCTACTGTAAATTGA ATTCCACCCACCATCTGTC C	Asuka <i>et al.</i> (2004)	NED	1.13	(AG) ₂₁	112–130	9	7.75	0.730	0.687	–0.061
csolfagus_05	GGTTTCTAGCAAAATTGGCAAT CCCAAAAAGGCCCTACTACAA	G. G. Vendramin, personal communication	NED	1.88	(GA) ₁₀	167–179	7	5.75	0.710	0.698	–0.018
FS1_15	TCAAAACCCAGTAAATTTCTCA GCCTCAATGAACCTCAAAAAC	Pastorelli <i>et al.</i> (2003)	PET	2.25	(GA) ₂₆	95–137	16	11.25	0.870	0.829	–0.049
sfc_0036	CATGCTTGACTGACTGTAAAGTTC TCCAGGCCTAAAAACATTTATAG	Asuka <i>et al.</i> (2004)	FAM	2.44	(TC) ₂₃	94–112	10	8.50	0.830	0.792	–0.049
csolfagus_06	GTGTGTGCTCACAGCAGTCG ACGCTTGCTCTCTTCTGCACT	G. G. Vendramin, personal communication	PET	1.88	(AG) ₁₃	203–221	9	8.00	0.867	0.809	–0.072
csolfagus_19	TGCCCATGAGGTTTGTATCA GCCGAATAAACCCAGAAAACA	G. G. Vendramin, personal communication	VIC	0.84	(TC) ₁₃	154–182	10	8.75	0.853	0.800	–0.067
csolfagus_29	CACAACTGCATTCCTTTC GTTGGCACCTTGGCTTGTT	G. G. Vendramin, personal communication	FAM	1.88	(CT) ₁₁	132–148	4	3.75	0.463	0.460	–0.008
<i>Kit-1</i> (mean)	–	–	–	–	–	–	9.3	7.50	0.760	0.723	–0.048
EEU75_A_0	TTCCAAACCAACCCCTTTATCC GACGGAGATTGAGGAAGAAACA	–	VIC	1.25	(CT) ₁₀	91–111	8	5.50	0.466	0.461	–0.011
DUKCT_A_0	GCCTCTGCAGCTCCTATAA GATCTAATGTGGGTTTGGTTTG	–	PET	4.50	(AC) ₁₄	75–95	6	4.75	0.727	0.680	–0.068
EJV8T_A_0	CCTGTCTCACACTTGGGTCTA TGCATTACAAAGCCTGAAACA	–	NED	2.25	(TC) ₁₀	143–155	6	5.00	0.606	0.620	0.022
EMILY_A_0	GACCCCAAGGTACAGTGCT CGTACAAATTGCACCCACATC	–	FAM	2.25	(GA) ₁₁	142–152	6	5.50	0.713	0.728	0.021
ERHBI_A_0	TGCAACAACTTAGCACCTTTGA GGGTGGCTTATCCAAAAT	–	PET	1.25	(AG) ₉	159–167	4	3.50	0.669	0.642	–0.042
DZ447_A_0	GGTGCAATCTTACCTTAGGACA ATAGGAGTGGGACGGCTAGG	–	NED	3.75	(TC) ₁₀	186–194	6	6.00	0.761	0.735	–0.036
concat14_A_0	TGAAGAAATTCACAAACCCCAACA GGGTGTTTACGATGGTGGA	–	VIC	1.13	(TC) ₉	173–197	5	4.75	0.814	0.752	–0.082
DE576_A_0	TCTCCTTAGATCCACAATCACA AGCTCTTCATTGCTCAGAACG	–	FAM	6.10	(CAA) ₁₀	211–232	7	5.75	0.790	0.707	–0.117
<i>Kit-2</i> (mean)	–	–	–	–	–	–	6.0	5.09	0.693	0.666	–0.039
Total (mean)	–	–	–	–	–	–	7.6	6.30	0.727	0.695	–0.044

[C], final concentration in each primer premix (μM); nA, total number of alleles; A, allelic richness; H_o, observed heterozygosity; H_e, expected heterozygosity; F_{IS}, fixation index.

cycles. PCR products were then separated using a 2% agarose gel stained with GelRed (Biotium, USA). For the subsequent genotyping step on ABI-3730 (Applied Biosystems, USA), PCR products were diluted in pure water depending on their intensity. As internal size standard we used LIZ600.

Microsatellites scoring

STRAND software was used to determine individual genotypes (Veterinary Genetics Laboratory, University of California; <http://www.vgl.ucdavis.edu/STRand>). To detect discrete size variants, allele binning was based on raw size using AUTOBIN (an Excel macro available at <http://www4.bordeaux-aquitaine.inra.fr/biogeco/Ressources/Logiciels/Autobin>; see Guichoux *et al.* 2011a).

Diversity analyses and error rate measurement

To assess genetic diversity over the four populations, the total number of alleles (n_A), allelic richness (A ; mean number of alleles per population), observed heterozygosity (H_o), expected heterozygosity (H_e) and the fixation index (F_{IS}) were calculated at each locus using GenAlEx 6 (Peakall & Smouse 2006). The difference between the two kits in terms of their capacity to detect polymorphism was assessed by comparing the means of each diversity parameters using Mann–Whitney nonparametric test in R (R Development Core Team 2011). F_{IS} was calculated at each locus and over all loci to measure departures from Hardy–Weinberg equilibrium (HWE). For each multiplex kit, linkage disequilibrium was assessed for each locus pair across all populations using GENEPOP 4.0 (Rousset 2010). False discovery rate (FDR; Benjamini & Yekutieli 2001) control was used to adjust P -values (to q -values) for multiple comparisons (i.e. tests for departure from HWE and linkage disequilibrium) to minimize type I errors (Storey 2003). FDR control was carried out using package qvalue (Dabney *et al.* 2011) in R (R Development Core Team 2011).

A first error estimate was achieved by counting the mismatches from a subset of 80 duplicated samples (from two beech stands; 48% of the complete dataset), each individual being re-genotyped at all loci. In a second time, we measured the disagreement rate between two readers for the 16 loci over 165 samples. We classified incoherencies as type A: the two readers selected different alleles for the same sample and as type B: an allele is detected by only one reader. Whenever a sample was genotyped differently by the two readers, we tried to identify a consensus genotype. When no consensus genotype could be determined, we set missing data.

Results and discussion

Multiplex PCR optimization

Starting with 31 preselected SSRs (14 from *kit-1* and 17 from *kit-2*), we excluded 15 (three with null alleles, three with low polymorphism (<4 alleles), six with poor amplification quality and three with low signal intensity). This led to a total of 16 markers multiplexed in two 8-plex kits (Table 2) showing clear profiles and homogeneous amplification. The analysis of the 10 families revealed no null allele at any of the 16 SSR loci.

All of the SSR markers had di-nucleotide repeats except DE576_A_0 (tri-nucleotide) (Table 2, Figs 1 and 2). Note, however, that one marker from *kit-2* had one off-ladder microvariant (DZ447_A_0, allele 187; Fig. 2). This allele was shown to segregate in the progeny test, thus it is not an amplification artefact. As the binning was performed using raw allele size, such an allele was readily identified. These results emphasize the importance of analysing samples using raw sizes and to bin the alleles subsequently (Guichoux *et al.* 2011b).

Once multiplexed, four of the 16 loci were found to exhibit incomplete adenylation which could compromise peak recognition (csolfagus_05, csolfagus_06, EJV8T_A_0, EMILY_A_0). Their profiles presented split peaks including the *A-* form (incompletely adenylated, blunt ended product) and *A+* form (1 bp longer, corresponding to the fully adenylated fragment) (Butler 2005). To promote full adenylation of the PCR fragment, the addition of a 'PIG-tail' sequence (5'-GTTTCTT-3' or 5'-GTTT-3') at the 5' end of the reverse primer (non-labelled; Brownstein *et al.* 1996; Guichoux *et al.* 2011a) has been implemented on one microsatellite marker (csolfagus_05). This 'PIG-tail' could not be applied on all the loci presenting incomplete adenylation because of the decrease in the PCR efficiency with such tailed primers. To score the profile of the three other incompletely adenylated loci (csolfagus_06, EJV8T_A_0, EMILY_A_0), clear and robust reading rules set-up across readers were used to genotype the fully adenylated *A+* form.

SSR properties

Across all 165 sampled individuals, four failed to amplify (2.4%), perhaps because of low DNA quality or technical difficulties. Among the remaining individuals, 2.2% of the individual $\times$ locus combination could not be scored, mostly with *kit-2*.

Loci from *kit-1* were more polymorphic than those from *kit-2* [mean number of alleles (n_A): 9.3 ± 3.4 for *kit-1* and 6.0 ± 1.2 for *kit-2*; P -value = 0.017; mean allelic richness (A): 7.5 ± 0.4 for *kit-1* and 5.1 ± 0.2 for *kit-2*; P -value = 0.015] (Table 2). There was no significant difference in

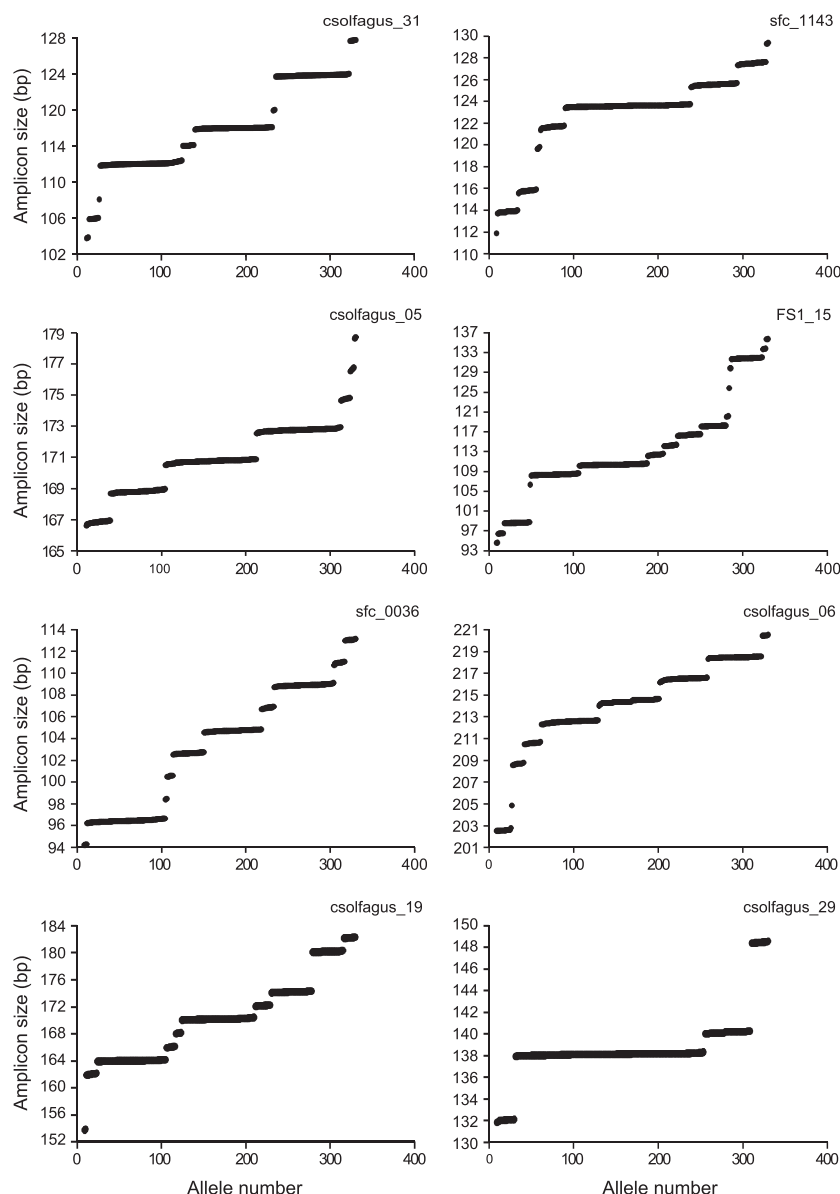


Fig. 1 Raw allele size distribution for the eight microsatellite markers included in multiplex *kit-1* developed on European beech (*Fagus sylvatica*).

observed heterozygosity (H_o) or fixation index (F_{IS}) between *kit-1* (0.76 ± 0.03 and -0.048 ± 0.023 , respectively) and *kit-2* (0.69 ± 0.02 and -0.039 ± 0.049 , respectively) (P -values = 0.195 and 0.713, respectively) (Table 2). The differences in polymorphism between the two kits (i.e. for parameters nA and A) could be partly related to the repeat number of the microsatellite motif: 16.1 on average for *kit-1* (ranging from 11 to 26 repeats) and 10.4 for *kit-2* (ranging from nine to 14 repeats; P -value = 0.014). SSR loci with greater number of repeats have higher mutation rates, therefore leading to higher levels of polymorphism (Ellegren 2000; Petit *et al.* 2005; Kelkar *et al.* 2008). Accordingly, van Asch *et al.* (2010)

suggest selecting SSR loci having more than 11 repeats. Several loci of *kit-2* are below this threshold. For this kit, screening identified 227 loci with an average read length of 142 bp (min and max length: 90 and 290 bp, respectively). Longer reads allowed by recent 454 technology development (>400 bp; Schuster 2008) should facilitate the identification of polymorphic SSRs by increasing the flexibility to design primers around the most promising SSRs. In addition, deeper coverage of the genomic DNA (our initial screening was limited to 1/32th of a 454 run) and more efficient enrichment procedures should help identify more candidates with sufficient repeat numbers.

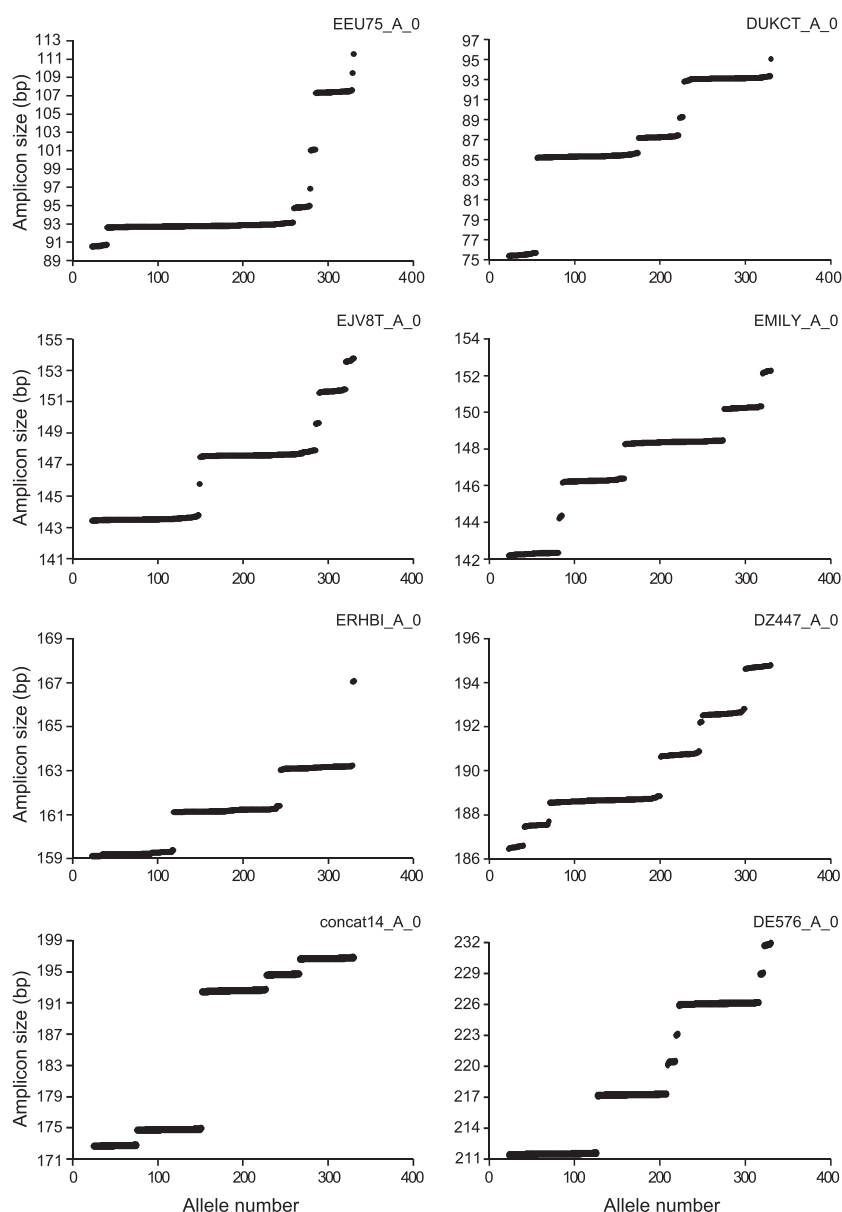


Fig. 2 Raw allele size distribution for the eight microsatellite markers included in multiplex *kit-2* developed on European beech (*Fagus sylvatica*).

Most loci have negative fixation index (F_{IS}), except two loci from *kit-2* (0.022 for EJV8T_A_0 and 0.021 for EMILY_A_0) (Table 2). The mean F_{IS} across all loci and populations is -0.044 (Table 2). Such slightly negative fixation indexes reflect excesses in heterozygotes. Despite this, no locus showed deviation from HWE after FDR control.

One locus pair, SFC_1143 and SFC_0036, was in linkage disequilibrium after control of the FDR (q -value = 0.036). Either these loci could be physically linked or the linkage disequilibrium could be caused by demographic factors such as bottleneck or immigration (Hedrick 2005). Note that linked markers can be useful to estimate gene

flow, introgression (Koopman *et al.* 2007) or population structure (Falush *et al.* 2003). The probability of identity (Paetkau *et al.* 1995) was calculated using *IDENTITY 4* (Wagner & Sefc 1999) after excluding one of the two loci in linkage disequilibrium (SFC_0036). We found an estimate of 2.1×10^{-8} , indicating that the probability of finding two randomly matching genotype is still extremely low.

Error rate measurement

The first error rate based on 80 duplicated genotypes yielded a mismatch rate of 0.08%, suggesting high

robustness of the 16 markers. The mean 'disagreement rates' between two human readers was 0.87% that is 10-fold higher. Most of the differences (91%) were because of Type B errors (typically caused by one reader failing to detect an allele), whereas wrong allele calling (Type A error) represented 9% of the mismatches. Most of the reading errors resulted from careless mistakes that could be easily corrected. Indeed, this kind of error was completely eliminated by defining clear and robust reading rules.

SSR transferability

The overall transferability of the 16 SSRs among the 27 individuals from seven other beech species is 92% (*kit-1*: 94%; *kit-2*: 90%). In *kit-1*, two loci failed to amplify in *Fagus engleriana* and *F. lucida*: csolfagus_05 and FS1_15. In *kit-2*, the marker DZ447_A_0 could be transferred only to *F. orientalis* and *F. taurica*, two species closely related to *F. sylvatica* (Denk *et al.* 2002). As expected, numerous new alleles were identified in these species: across the seven species, the number of new alleles ranged from two to 12 for *kit-1* and from one to 12 for *kit-2* (Table S1, Supporting Information). In our transferability assays, amplification was not always successful and profiles were heterogeneous which might be a result of low DNA quality or technical difficulties. Therefore, Mendelian segregation analyses and further amplification tests on large populations will have to be carried out before concluding that these markers can be successfully transferred to these species.

Conclusion

These two multiplex kits of SSR markers should be useful for future population genetics studies on *Fagus sylvatica* by providing powerful and accurate genotyping tools. We are currently genotyping >2000 individual beech trees using the two kits, thereby demonstrating the effectiveness of this approach for large-scale genotyping surveys. Moreover, these markers were efficiently transferred to other *Fagus* species, albeit additional tests and quality controls should be performed on larger samples.

Authors' Contributions

RJP and SL conceived the study. SL performed the experiments and produced the data. SW analysed the 454 data and designed primers with QDD software. SL and GdL did the genotyping. GdL analysed the data (Meglécz *et al.* 2010). SL wrote the paper with the help of GdL, SW, and RJP. All authors have read and approved the manuscript.

Acknowledgements

We thank Alexis Ducouso, Erwan Guichoux, and Lélia Lagache for discussion and support during various steps of the project. The following colleagues kindly provided information on primers used in *kit-1* GG Vendramin (CNR, Italy), B Fady and S Odou-Muratorio (INRA, France) and B Vornam, S Seifert and R Finkeldey (Univ. Göttingen, Germany). We thank Emmanuel Corcket, Jean-Marc Louvet and Elodie Magnanou for help in the field and for providing biological material, Guillaume Lalanne-Tisé and Patrick Léger for laboratory support, Christophe Boury for developing robotic applications used in this project. Yves de Lafontaine, Dominique Piou, Nicolas Perrette, and Nobuhiro Tomaru provided biological material from other *Fagus* species. Genotyping was performed in the Genome-Transcriptome facility of the Functional Genomic Center of Bordeaux with the help of Franck Salin and Sarah Monllor. Experiments were funded by Ministère de l'Agriculture et de la Pêche, DGPAAT Convention E30/08 to Alexis Ducouso. GdL benefited from a 'Fonds québécois de la recherche sur la nature et les technologies' (FQRNT) postdoctoral fellowship.

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Data Accessibility

Microsatellite data were deposited in the DRYAD repository doi:10.5061/dryad.pc8d30j2.

Supporting Information

Additional supporting information may be found in the online version of this article.

Table S1 Details of the transferability of the two multiplex kits on seven other *Fagus* species.

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